This article has been accepted for publication in IEEE Transactions on Geoscience and Remote Sensing. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TGRS.2026.3737791



> TGRS-2026-02307 <

# Unveiling profile distribution of soil organic carbon in China's largest freshwater wetland using a novel 3D INLA-SPDE model by integrating remote sensing and ground field survey data

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***Abstract*—Three-dimensional (3D) digital soil mapping can quantify the horizontal and vertical variations of soil properties. However, most previous studies on the 3D distribution of soil properties mainly relied on geostatistical methods or hierarchical predictions generated by machine learning methods, which often suffer from problems such as low computational efficiency, poor estimation accuracy, insufficient quantification of prediction uncertainty, or black box effects. To fill these gaps, this study developed an efficient and accurate 3D spatial prediction model named three-dimensional (3D) Integrated Nested Laplace Approximation (INLA) and Stochastic Partial Differential Equation (SPDE) model (3D INLA-SPDE) for soil organic carbon (SOC) by combining multi-source remote sensing data with ground observations. Taking Poyang Lake Wetland, China's largest freshwater wetland, as the study area, we developed a star–ground data fusion framework. This framework integrates 1270 soil profile samples (0–100 cm depth) with remote sensing datasets, such as terrain indices and vegetation metrics, to characterize the 3D spatial variability of SOC. Key findings reveal a pronounced depth-dependent decline in SOC content (7.34 g/kg in surface layers vs. 2.35 g/kg in deepest layers), with high-value zones concentrated in the southeastern wetland. Soil properties and topographic factors emerge as primary drivers of 3D SOC variation. The proposed 3D INLA-SPDE model outperforms conventional geostatistical and machine learning approaches in prediction accuracy ($R^2$ improved by 77.78%–116.22%, RMSE reduced by 40.23%–44.68%), computational efficiency (37.5% faster processing), and uncertainty representation. By generating high-resolution SOC spatial maps, this model advances scientific understanding of 3D SOC distribution patterns and regulatory mechanisms in wetlands. The results could directly support carbon sequestration accounting and climate-smart wetland management, offering a novel tool for 3D digital soil mapping.**

***Index Terms*—3D INLA-SPDE, soil organic carbon, digital soil mapping, computational efficiency, wetland.**

## I. INTRODUCTION

SOIL ORGANIC CARBON (SOC) is a core component of soil fertility and ecosystem functioning, playing a vital role in the global carbon cycle [1], [2], [3]. A comprehensive understanding of its three-dimensional (3D) spatial distribution of SOC is therefore critical for advancing sustainable wetland management and climate change mitigation strategies [7], [8].

Wetlands are pivotal in this context, constituting a critical yet vulnerable component of the global carbon pool. Although covering only about 6% of the Earth's land surface, wetland is estimated to store approximately one-third of global SOC [4], [5], [6]. However, the processes governing SOC accumulation and decomposition in wetlands are exceptionally complex [9], [10]. Dynamic hydrological conditions, coupled with intensifying anthropogenic pressures and climate change, drive strong spatial heterogeneity in SOC stocks (SOCS), resulting in intricate 3D distribution patterns that are challenging to quantify [11].

Accurately mapping 3D variation of SOC requires robust spatial prediction methods. While significant progress has been made in 3D digital soil mapping [12], [13], [14], [15], [16], [17], prevalent approaches are often constrained by inherent limitations. Traditional geostatistical methods, such as 3D Empirical Bayesian Kriging (3D EBK), explicitly account for spatial autocorrelation but can be computationally inefficient and sensitive to parameter selection, especially with large datasets [2], [18], [21]. Conversely, machine learning techniques like the hierarchical 3D Random Forest (3D RF) model effectively capture non-linear relationships but are often criticized as "black-box" models due to their poor interpretability and limited capacity for quantifying prediction uncertainty [22], [23]. Simpler methods like 3D Inverse

This study was supported by the National Science Foundation of China (No. 42201073), the Jiangxi "Double Thousand plan" (No. jxsq202301091), the Natural Science Foundation of Jiangxi Province (No. 20232BAB213058), the National Science Foundation of China (No. 42471060), and International Scientific Research Seed Fund of JUFE. (Corresponding author: Modian Xie.)

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Distance Weighting (3D IDW), while easy to apply, rely on the assumption of spatial isotropy and can produce artifacts with low accuracy, particularly with unevenly distributed data [16], [18], [20].

To address these limitations, this study develops and applies an innovative 3D spatial modeling framework based on the Integrated Nested Laplace Approximation (INLA) with Stochastic Partial Differential Equations (SPDE) (3D INLA-SPDE). This model integrates the advantages of Bayesian and spatial statistics within a computationally efficient framework. Specifically, it employs an SPDE to represent 3D spatial dependence, discretizes the domain using a triangular mesh, and approximates the solution via a Gaussian Markov Random Field (GMRF) for sparse matrix operations that reduce computational burden [24], [25], [26], [27], [28]. Crucially, it uses INLA for fast and deterministic Bayesian inference, avoiding the convergence issues of traditional Markov Chain Monte Carlo (MCMC) methods [29]. Another key advantage of our 3D INLA-SPDE model is its ability to provide a full posterior distribution, enabling rigorous quantification and visualization of prediction uncertainty [30].

To validate this framework, we implemented it in China's largest freshwater wetland, the Poyang Lake Wetland, integrating 1270 soil profile samples (0–100 cm depth) with 38 environmental covariates. The specific objectives of this study are: (1) to evaluate the performance of the 3D INLA-SPDE model against established 3D mapping methods (3D IDW, 3D EBK, and 3D RF); (2) to reveal the horizontal and vertical distribution patterns of SOC in the Poyang Lake wetland; and (3) to identify the key environmental factors controlling the 3D variation of SOC. This research aims to deepen the mechanistic understanding of wetland SOC dynamics and advance the application of robust 3D soil mapping technologies in complex ecosystems.

## II. Materials and Methods

### A. Study Area

The Poyang Lake Wetland (28°11'~29°51' N, 115°31'~117°06' E), located in northern Jiangxi Province, China, represents the country's largest freshwater wetland system within the middle reaches of the Yangtze River (Fig. 1). This wetland experiences a pronounced subtropical humid monsoon climate, which drives a dramatic seasonal hydrological cycle [31], [32], [33], [34]. The humid season (May to October) is characterized by flooding and substantial sediment deposition across the lake basin. Conversely, during the dry season (November to April), water recedes, exposing extensive areas of the lake bed. This cyclical shift between inundation and exposure creates dynamic and heterogeneous environmental conditions that are known to strongly influence SOC accumulation, decomposition, and overall spatial distribution [31], [35], [36]. The resulting complex 3D heterogeneity in SOC makes this region a critical and challenging area for high-resolution soil carbon mapping, and pose a great challenge to climate-smart wetland management.

**Fig. 1.** Spatial relative position of the sampling network within the study watershed.

### B. Data Collection

*1) Soil Data Collection:* The soil survey was conducted from December 2023 to January 2024 across the Poyang Lake Wetland, covering an area of 2640.29 km². The soil samples were collected following a 4 km × 4 km grid-based random sampling scheme. Minor adjustments to the preset sampling locations were made during field work to account for accessibility. In total, 1270 soil samples were collected from 127 sites with a maximum a depth of 100 cm using a Rhino S1 soil sampling drill. The geographic coordinates of each sampling point were recorded with a handheld GPS. Each soil sample was subdivided at 10 cm depth intervals, resulting in 10 subsamples per profile. Soil organic matter (SOM) content was determined via the potassium dichromate oxidation method after air-drying the samples and grinding them to pass through a 2-mm sieve. The SOM measurements were then converted to SOC content using the Bemmelen conversion factor of 0.58 referring to numerous previous studies [37], [38], [39].

*2) Covariate Dataset and Preprocess:* A total of 38 environmental covariates representing soil properties, climate, topography, and biology were collected to predict 3D variation of the SOC in the Poyang Lake Wetland (Table I). All original environmental covariate raster datasets were reprojected to the WGS 1984 coordinate system. Discrete data were resampled using the nearest-neighbor interpolation method, while continuous data were processed with bilinear interpolation, achieving a unified spatial resolution of 90 m for all raster data. Finally, the spatial extent of all covariates was clipped to match the boundary of the Poyang Lake Wetland. All covariates were then standardized to ensure dimensional homogeneity for subsequent analysis. Detailed sources and descriptions of all covariates are provided in the Supplementary Materials.

**TABLE I**
LIST OF COLLECTED COVARIATES

*3) Covariate Selection:* We applied the Recursive Feature Elimination (RFE) algorithm to filter the optimal covariate dataset [40]. Model predictive performance was evaluated using a 10-fold cross-validation approach [41]. Variable significance was evaluated iteratively by refitting models with different covariate combinations. In each iteration, the least significant covariates were removed, and the model was refitted. The final feature set was determined by selecting covariates from the model achieving the smallest root mean square error (RMSE) [40], [41].

### C. Methods

*1) Latin Hypercubic Sampling:* Latin Hypercube Sampling (LHS) is a stratified sampling method that provides a more reliable characterization of parameter changes compared to simple random sampling, leading to more robust estimation results [42], [43]. It employs a comprehensive random stratified strategy to effectively cover the entire parameter space, making it well-suited for handling numerous predictor variables and heterogeneous landscapes [44], [45], [46]. The LHS method is based on the inverse-function transformation technique. It include the following steps: First, as a stratified sampling method, it ensures that the sampled values cover the entire distribution range of the input random variables. This improves sampling efficiency. Suppose there are $N$ samples and $K$ input

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random variables. Let the cumulative distribution function of the variables be $X_k$. Divide the distribution function into $N$ sub-intervals within the range. Draw a random number uniformly from each sub-interval. Then, it obtain the corresponding values from the inverse function. Each sub-interval is sampled only once to avoid duplication. Currently, the main approach used for LHS is Median Latin Hypercube Sampling. In this method, the midpoint of each interval is extracted as a sample in each step. After that, the $n$-th ($n$=1, 2,⋯, $N$) sampling value is obtained from the inverse function to form an $K$-dimensional sequence:

$$X_{k,n} = F_K^{-1}\left(\frac{n-0.5}{N}\right) \quad (1)$$

Where $N$ is the number of samples and $K$ is the number of input variables.

*2) Three-dimensional Inverse Distance Weighting:* IDW is a highly precise, universally applicable spatial interpolator, as initially validated by Watson and Philip [47]. It has become a prevalent tool in spatial estimation endeavours, as evidenced by its extensive utilization in various studies [48]. It estimates the value at an unsampled location by calculating a distance-weighted average of values from neighboring sample points. The weight assigned to each sample point decreases as its distance from the prediction location increases. The conventional formulation for the IDW method is as follows:

$$D(a) = \sum_{x=1}^{y} w_x D(a_x) \quad (2)$$

where $D(a)$ represents the predicted value of the interpolation point; $D(a_x)$ ($x$ = 1, 2, ... $y$) represents the observed value of the $x$-th sampling point; y is the number of observations near the interpolation point; $w_x$ is the weight assigned to the $x$-th sampling point, which is usually calculated as:

$$w_x = \frac{1/d_x^p}{\sum_{x=1}^{y} 1/d_x^p} \quad (3)$$

Where $d_x$ represents the Euclidean distance measured from the interpolation point to the $x$-th sampling point. The parameter $p$ is designated as the distance decay parameter, which governs the rate of weight reduction as the distance increases. In the realm of IDW applications, the value of $p$= 2 is most frequently adopted [49]. The IDW prediction is contingent on several parameters, including the number of neighboring samples $n$, the Euclidean distance $d$, and the distance-weighting coefficient $p$. Because it assumes spatial isotropy, this method is straightforward to apply. The 3D IDW extends the classic inverse distance weighting to 3D space, combining vertical ($z$) and horizontal ($x, y$) coordinates to perform continuous attribute interpolation from discrete points. The interpolation of the target position is the weighted average of adjacent observations, with weights inversely proportional to the Euclidean distance and raised to the parameter $p$, which controls the weight attenuation. The implementation includes preprocessing 3D point datasets, generating voxel meshes, and adopting a search neighborhood strategy.

*3) Three-dimensional Empirical Bayesian Kriging:* Based on Bayesian theory, Bayesian Kriging's model parameters are estimated by the posterior distribution [50]:

$$p(Z|\theta) \propto f(Z|\theta)p(\theta) \quad (4)$$

where $p(Z|\theta)$ is the likelihood function, representing the posterior probability distribution; $p(\theta)$ is the prior probability distribution; $Z$ is the observed attribute value; the vector of the $\theta$ variogram model parameters; $f(Z|\theta)$ depends on the maximum likelihood function of other continuous functions. Assuming that the observed attribute value $Z$ is a common dispersion with a coefficient $\beta$, and the variance of the data is $\sigma^2$. The 3D-EBK algorithm extends this framework into 3D space, providing an efficient solution for automated interpolation. A key advantage of 3D-EBK over conventional kriging is its subset-processing mechanism, which autonomously derives semivariogram parameters, eliminating the need for manual tuning [51], [52]. Furthermore, unlike the traditional kriging model, the 3D EBK model compensates for errors arising from the estimation of the underlying semi-variogram [4].

4) *Three-dimensional Random Forest:* As described by Breiman [53], the RF model is a machine-learning approach that aggregates decision trees. In RF, the final prediction is obtained by aggregating the predictions from each individual tree in the ensemble. The RF model offers several strengths including mitigating overfitting, handling high-dimensional data efficiently, and supporting parallel operation. As a result, the RF model has been increasingly applied in Digital Soil Mapping (DSM), where it effectively captures non-linear relationships among variables [41], [54], [55]. Moreover, it excels at addressing modeling challenges characterized by high-dimensional covariate spaces, primarily due to its strong resistance to overfitting [53], [56], [57], [58]. Two essential parameters of RF model are "*mtry*" and "*ntree*". To determine the optimal hyperparameter configuration for the RF algorithm, 10-fold cross-validation was used in the present investigation. Specifically, we set "*ntree*" to 500 and "*mtry*" to 5.

To predict the 3D spatial distribution of SOC, we employed a hierarchical RF modeling strategy. A separate RF model was constructed and trained for each soil layer using a portion of the data. The model's performance for each layer was validated internally using the out-of-bag (OOB) error estimate, which provides an unbiased assessment and allows for parameter fine-tuning. Finally, the predictions from all stratified models were integrated to generate a complete 3D spatial distribution map, ensuring the final output accurately reflects SOC variability across different depths and locations within the study area.

5) *Three-dimensional Integrated Nested Laplace Approximation-Stochastic Partial Differential Equation:* The analysis employed the INLA-SPDE approach, implemented in the *R-INLA* package, building upon the methodological framework of Cameletti et al. [59] and Rue et al. [60]. The INLA implementation involves four main stages: (1) constructing a Delaunay triangulation mesh, (2) defining the SPDE model, (3) assembling the data structure, and (4) performing Bayesian inference and estimation. The construction of the computation mesh is crucial for model performance. It is built using constrained Delaunay triangulation, with initial vertices at observation locations. Mesh quality is controlled by three key parameters. The first parameter is the *offset*, which defines the mesh extension distance to minimize boundary effects. The second parameter is the *cutoff,* which specifies the minimum allowed distance between vertices to avoid overly small triangles. The third parameter is the *maxedge*, which sets the maximum allowable edge length for triangles. Together, these parameters govern the trade-off between Gaussian Markov Random Field (GMRF)

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approximation accuracy and computational cost. Both accuracy and cost depend on the number of mesh vertices: more vertices yield finer GMRF approximations but incur higher computational cost [61]. However, prediction accuracy is highly sensitive to the selection of truncation parameters. Specifically, if the truncation value is too large, the model fails to capture small-scale spatial autocorrelations [62]. For spatial domain extension, Lindgren and Rue [27] recommend that the extension distance should exceed the variogram range. Conversely, Fuglstad et al. [63], [64] suggest that the maximum edge length should be smaller than the spatial correlation range. Based on these considerations, we set the *offset* to 6 km, *cutoff* to 150 m, and *maxedge* to 6 km. A comprehensive description of the INLA-SPDE model is provided by Sun et al. [30].

In this study, we extended the INLA-SPDE spatial prediction framework from two dimensional space to 3D space. Using the profile SOC data collected from the Poyang Lake Wetland, we established a 3D model based on the INLA-SPDE framework. For fully observed data, the model configuration is explained as follows:

$$y(s_i, d) = z(s_i, d)\beta + \xi(s_i, d) + \varepsilon(s_i, d), \varepsilon \sim N(0, \sigma_\varepsilon^2) \quad (5)$$

where $y(s_i, d)$ represents the response data of a certain planar location $s_i$and depth $d$, $z$ represents the vector of the $z$ covariate, $\beta$represents the vector of the corresponding coefficients of the covariate, $\xi$ represents the realization of the so-called state process (the soil change process that was not truly observed in this study), $\varepsilon$ represents the Gaussian white noise process of the measurement error, with an average value of 0. The variance is $\sigma_\varepsilon^2$. $\varepsilon$ is a 3D Gaussian field (GF) (autoregressive model) that varies with depth, with a coefficient of a and an innovation amount of $x$ that is spatially dependent [65]. For detailed information about the model, please refer to the supplementary materials.

*6) Model Accuracy Evaluation:* To validate the 3D model, the 127 profiles containing 1270 soil samples were divided into training (102 profiles, n = 1020) and validation (25 profiles, n = 250) datasets using LHS with an 8:2 ratio. The optimal feature set was selected using the training dataset, and the model was subsequently trained. The model's spatial prediction performance was evaluated on the validation dataset. Two extensive adopt metrics were used for model's performance evaluation: the coefficient of determination ($R^2$) and the RMSE [41], [66], [67], [68]. Their formulas are as follows:

$$R^2 = 1 - \frac{\sum_i^n (y_i - x_i)^2}{\sum_i^n (y_i - \bar{y})^2} \quad (6)$$

$$RMSE = \sqrt{\frac{\sum_i^n (y_i - x_i)^2}{n}} \quad (7)$$

where $n$ denotes the quantity of samples; $y_i$ and $x_i$ respectively denote the observation and predictive SOC values; $\bar{y}$ and $\bar{x}$ respectively denote the SOC averaged observation and predictive values.

### *D. Estimation of Soil Organic Carbon Storage*

In this study, we estimated the SOCS in Poyang Lake Wetland within 100 cm soil profiles. Since $BD_i$ was not measured, we used the empirical formula reported by Sun et al. [34] to estimate the value of $BD_i$. The formula is as follows:

$$BD_i = 1.0471 exp\left(\frac{-SOC}{14.4343}\right) + 0.5230 \quad (8)$$

The formula for determining the SOC density (SOCD) within the research area is articulated as follows:

$$SOCD_i = \sum_{i=1}^n SOC_i \times BD_i \times H_i \quad (9)$$

The formula for determining the SOCS within the research area is articulated as follows:

$$SOCS_i = \sum_{i=1}^n SOCD_i \times r_i^2 \times (1 - CF_i) \times 10 \quad (10)$$

where $SOCD_i$ represents SOCD (kg $m^{-2}$); $BD_i$ is the soil bulk density(g $cm^{-3}$); $H_i$ is the sampling depth (m); $SOCS_i$ is SOCS (Tg C); $r$ is the edge length of the cell; $CF_i$ is the value of coarse fragments (%).

### *E. Explainable Machine Learning*

The Shapley Additive Explanation (SHAP) was used to quantify the importance of different variables and identify the main controls of 3D variation of SOC. Rooted in cooperative game theory's Shapley value, the SHAP method provides a unified framework for interpretability model [69], [70]. It quantifies the marginal contribution of each feature to the final prediction. Detailed theoretical foundations of SHAP values are described by Padarian et al. [71]. We generated SHAP partial dependence plots (PDPs) for all environmental covariates used in the predictive model. These plots were used to visualize the direction and magnitude of the effect of each covariate on SOC accumulation.

### *F. Uncertainty Analysis*

In the 3D INLA-SPDE model, uncertainty quantification is directly based on the posterior edge distribution of the potential Gaussian field. For each prediction node in the 3D space, the INLA algorithm not only calculates the mean (or median) of the posterior prediction distribution to represent the best estimate of soil organic carbon content, but also calculates the key statistics that characterize the prediction accuracy. Specifically, the standard deviation (or variance) of the posterior edge distribution at each 3D spatial position is utilized to quantify the uncertainty of the local prediction results. Furthermore, based on the posterior prediction results of the 3D INLA-SPDE model, this study extracted the 95% confidence interval (i.e., the 2.5% and 97.5% quantiles of the posterior distribution), thereby providing a probability measure of the prediction range. These uncertainty indicators are visualized throughout the 3D space of the entire wetland to identify areas with lower predictive reliability. This explicit uncertainty output provides a more rigorous basis for risk assessment and decision-making in soil carbon mapping and monitoring [30], [72].

### *G. Statistical Analysis*

All statistical analyses, including LHS, covariate selection, and SOC prediction, were performed using R version 4.3.3. Optimal feature selection was implemented using the "*rfe*" algorithm from the "*caret*" package. The 3D INLA-SPDE model was developed with the "INLA" package [72]. SHAP values were calculated on the training dataset using the "*kernelshap*" package [69], [73]. The spatial distributions predicted by the 3D IDW, 3D EBK, and 3D RF models were generated and mapped using ArcGIS Pro 3.4.3. All raster data processing was conducted using ArcGIS Pro 3.4.3 and SAGA GIS 9.2. The workflow of this study is shown in Fig. 2.

**Fig. 2.** Workflow of the present study.

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## III. Results

### A. Descriptive Statistics of SOC

The descriptive statistics of SOC in different soil layers in Poyang Lake Wetland are presented in Table II. The mean values and median values of SOC in the whole dataset, training and validation dataset in different soil layers are very close although clear differences were observed on the maximum values due to existence of several extreme values (Table II). In addition, a significant and consistent decreasing trend in SOC content was observed with increasing soil depth (Table II, Fig. 3).

**Fig. 3.** SOC depth curve graph.

**TABLE II**
DESCRIPTIVE STATISTICS OF SOC (G/KG) AT DIFFERENT SOIL DEPTHS (CM)

### B. Featured Variable Selection

As shown in Fig. 4, the RMSE of cross-validation generally fluctuates up and down with the increase of covariates. The RFE algorithm achieved the lowest RMSE (2.3875 g/kg) when 30 covariates were used. Consequently, 30 covariates (Table III) were selected to develop the 3D-RF and 3D INLA-SPDE models for SOC prediction. Among them, 13 covariates (Sand, AK, Silt, pH, TK, AN, SP, SG, AP, CEC, Clay, TP, TN) are related to soil properties, and 6 covariates (TE, MAP, SAT, SR, MAT, SMC) are related to climate. Eight covariates (Slope, PC, MrVBF, TWI, VD, Aspect, LSF, TPI) are related to terrain, while three covariates (NDVI, EVI, NPP) are related to biology.

**Fig. 4.** Impact of the number of covariates on RMSE in RFE algorithm modeling.

**TABLE III**
OPTIMAL COVARIATES SELECTED VIA RFE ALGORITHM

### C. Assessment of Model Performance

Our results revealed that the 3D INLA-SPDE model obviously outperformed 3D IDW, 3D EBK, and 3D RF model with the highest predictive accuracy ($R^2$=0.80) (Fig. 5) and the lowest RMSE (1.04 g/kg), indicating minimal prediction error. In contrast, the 3D IDW ($R^2$= 0.37), 3D EBK ($R^2$= 0.45), and 3D RF ($R^2$= 0.40) models showed lower $R^2$ values and higher RMSE values (1.85, 1.74, and 1.88 g/kg, respectively), indicating comparatively weaker predictive performance. Compared with 3D IDW, 3D EBK, and 3D RF, the $R^2$ of 3D INLA-SPDE increased by 116.22%, 77.78%, 100%, while the RMSE reduced by 43.78%, 40.23%, 44.68%, respectively. Overall, the 3D INLA-SPDE model demonstrated superior performance compared with existing widely used methods.

**Fig. 5.** Scatter plots of SOC in the external validation dataset and the corresponding mapping accuracy results (black dotted lines represent 1:1 fitting lines): (a) 3D IDW method; (b) 3D EBK method; (c) 3D RF method; (d) 3D INLA-SPDE method.

### D. 3D Distribution Patterns and Stock of SOC

Our results indicated that the 3D IDW, 3D EBK, 3D RF and 3D INLA-SPDE model revealed similar 3D distribution patterns of SOC in general (Fig. 6). However, compared with 3D IDW, 3D EBK, and 3D RF, 3D INLA-SPDE can capture more detailed pattern of SOC in the Poyang Lake Wetland (Fig. 6). All the models revealed a pronounced decline in SOC content with increasing depth. Specifically, the SOC content decrease from 0.34~8.18 g/kg in the 0~10 cm layer to 0.19~4.12 g/kg below 30 cm depth. Areas with high SOC content (>6.0 g/kg) were predominantly clustered in the western, central, and eastern regions of the study area. The extent of the area with high SOC values shrinks with increasing soil depth. In contrast, areas with low SOC content (<4.0 g/kg) were majorly distributed in the fringe and southern parts of the study area.

**Fig. 6.** Spatial distribution of SOC at each depth for each method at a resolution of 90 m: 3D IDW method; 3D EBK method; 3D RF method; 3D INLA-SPDE method.

In addition, in order to understand the current status of carbon sequestration in Poyang Lake Wetland, we also estimated the SOCS based on the SOC maps produced by different methods. As listed in Table IV, the SOCS estimated by different method varied between 92.24 and 114.83 Tg. These discrepancies likely originate from the different ways the methods handle spatial autocorrelation and data distributions (Table IV). Nevertheless, all models consistently highlighted the predominance of surface-layer SOC accumulation and the fragmented distribution of high-SOC regions in deeper layers. Considering the prediction accuracy of different models, especially the clear higher prediction accuracy of 3D INLA-SPDE, we have reasonable grounds to suggest that the estimated SOCS derived from 3D EBK, 3D IDW and 3D RF are overestimated. The 3D INLA-SPDE approach demonstrated robustness in capturing detailed horizonal and vertical patterns, providing valuable insights for wetland carbon management and soil conservation.

**TABLE IV**
SOCS FORECASTS THE RESULTS OF ALL MODELS

### E. The Contribution of Different Environmental Covariates to 3D Mapping of SOC

Using SHAP method, we quantified the relative contributions of different environmental covariates to mapping 3D distribution of SOC (Fig. 7). The results revealed that Sand was the most influential variable for 3D spatial prediction of SOC, followed by AK, Silt, and pH. Soil properties accounted for the largest proportion of explanatory power (53.7%) in mapping SOC (Fig. 7a), followed by topography (23.0%), climate (12.1%), and biological factors (11.2%). Based on the SHAP global contribution analysis (Fig. 7b), the relative importance of covariate categories for 3D SOC variation was: soil properties (0.594) > topography (0.255) > climate (0.134) > biology (0.124). The top ten variables by mean absolute SHAP value were: Sand (0.085) > AK (0.066) > Silt (0.064) > pH (0.059) > TK (0.057) > AN (AN, 0.057) > NDVI (0.046) > Slope (0.046) > EVI (0.046) > SP (0.042).

**Fig. 7.** The variable importance map sorts the SHAP values of the covariates in the 0~100 cm SOC 3D INLA-SPDE model. Each point on the graph represents the marginal contribution of a specific feature to the local prediction relative to the overall average prediction (indicated by the SHAP value of g/kg on the Y-axis of the dependency graph). Features are organized based on their average absolute contribution across all data points. Use different colors to represent the different feature values

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associated with each SHAP value.

## IV. Discussion

### *A. Mapping 3D Distribution of SOC through 3D INLA-SPDE Model*

In this study, we developed a novel 3D INLA-SPDE model to predict the spatial distribution of SOC in soil profiles of Poyang Lake Wetland and characterized its posterior parameters to enhance the interpretability of its results. The computational implementation utilized a system with an Intel Core i7-8750H CPU (6 cores @ 2.20 GHz), requiring around 50 minutes for model execution. The mesh constructed using the 3D INLA-SPDE framework (Fig. 8) successfully captured the spatial heterogeneity of 3D SOC distribution.

**Fig. 8.** The mesh constructed for the INLA-SPDE model. Pink indicates the study area boundary, and black dots indicate the sampling positions.

The posterior mean of the autoregressive parameter was 0.78 (Table V), indicating strong depth dependence in SOC content. The parameter exhibited a standard deviation of 0.016 and a narrow quantile range (0.855~0.917), suggesting that the comprehensive depth sampling resulted in low uncertainty in the dynamic predictions. This finding contrasted with results reported by Cameletti et al. [59], underscoring the critical role of detailed depth monitoring in enhancing 3D model performance.

**TABLE V**
POSTERIOR ESTIMATES (MEAN, STANDARD DEVIATION AND QUANTILE) OF FIXED EFFECTS, AND $\sigma_\varepsilon^2$, $\sigma_x^2$, *Range*, AND *a* IN THE 3D SOC MODEL

Note: *Range* represents the distance when the spatial correlation decays to nearly zero in a spatial random field. $\sigma_\varepsilon^2$ is the variance of the model; $\sigma_x^2$ is the Matérn variogram.

The posterior estimates of the spatial random effects indicated that the model variance $(\sigma_\varepsilon^2)$ had a mean of 0.310, while the Matérn variance $(\sigma_x^2)$ had a mean of 0.360 (Table V). The ratio between them was close to 0.86, indicating moderate spatial autocorrelation in SOC content. The mean of the spatial correlation range (Range) is around 4578 m (Table V), which differs in magnitude from the 44000~46000 m range reported by Sun et al. [74]. The specific numerical differences likely stem from substantial differences in spatial scale and topographic relief. The soil survey of Sun et al. [30] was conducted in Jiangsu Province with area of 107000 km$^2$, which has clearly larger spatial scale and thus would tend to bring about a larger spatial correlation range. On the other hand, Jiangsu Province is featured by uniform flat plain terrain which would result in weaker spatial variability of SOC and thus lead to a larger spatial correlation range [75]. It is worth noting that the standard deviation of the range parameter was 636 m, and its quantile range was relatively narrow (from 3466 to 5968 m). This confirms the high precision of the estimated spatial autocorrelation structure, which is attributed to the dense meshes in the study area (Fig. 8).

However, several issues still deserve attention. On the one hand, the sparse and uneven distribution of 2D spatial data leads to a relatively large standard deviation in the autoregressive process's parameters. On the other hand, although the estimates of spatial parameters are robust (e.g., the narrow quantile ranges of range and $\sigma_\varepsilon^2$), the unexplained variance $(\sigma_x^2)$ still accounts for a certain proportion, suggesting the possible existence of local variation factors not captured by the model. However, the standard deviations of the fixed-effect coefficients and the dense depth-sampling strategy jointly ensure the model's overall robustness.

### *B. Uncertainty Analysis of 3D INLA-SPDE Model*

We mapped the 95% confidence interval of the prediction for SOC to present the uncertainty of the SOC maps produced by 3D INLA-SPDE model within depth of 100 cm. As shown in Fig. 9, the predicted uncertainty displayed distinct patterns across the study area. Areas with high uncertainty were frequently located at the edges of the study area. This mainly derived from two reasons. On the one hand, usually, there are fewer sample points at the edges of the study area, and the lack of constraints from neighboring observations during spatial interpolation or extrapolation leads to increased prediction variance. On the other hand, the SPDE method relies on the spatial covariance structure, and at the boundaries, it cannot utilize "symmetric neighborhood" information, naturally resulting in higher uncertainty [76].

Additionally, uncertainty varied substantially with soil depth (Fig. 9). The top three soil layers (≤30 cm) exhibited higher uncertainty than deeper layers (>30 cm). This pattern is likely associated with the lower absolute SOC content in deeper soil layers (Table II). In shallow soil layers (0~30 cm), strong influences from dynamic surface environmental factors and anthropogenic activities contributed to complex spatial patterns, leading to relatively high uncertainty. Deep soil layers are less directly influenced by surface conditions and anthropogenic interference. Therefore, they exhibit relatively smaller uncertainty compared with top soil layers. Future improvements should enhance the model's capacity to simulate complex soil dynamic process in different soil layers [77], [78].

Overall, the prediction uncertainties of the SOC maps mainly originated from multiple sources. Primary among these were data limitations [79], [80]. Sampling deep soil layers is challenging, resulting in sparse data that inadequately represent subsurface conditions. This directly constrains the model's capacity to accurately characterize deep soil layers. For example, at 90~100 cm depth, data scarcity impedes accurate simulation of long-term pedological processes, leading to higher uncertainty. Model structural simplification was another key source of uncertainty [81]. The soil system exhibits high complexity due to interacting factors including climate, vegetation, topography, soil type, and anthropogenic activities. The model cannot fully incorporate all these complexities, inevitably reducing prediction accuracy.

**Fig. 9.** The results of uncertainty analysis of the 3D INLA-SPDE model.

### *C. Methodological Implications: Advantages of 3D INLA-SPDE*

In 3D continuous-space prediction, both computational efficiency and mapping accuracy are critical considerations. This study compared the 3D INLA-SPDE approach with two traditional geostatistical methods (3D IDW, 3D EBK) and a

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machine learning method (3D RF), revealing significant differences in computational performance and predictive accuracy. The 3D INLA-SPDE method achieved a substantially higher $R^2$ (0.80) compared to the traditional methods ($R^2$: 0.37~0.45) (Fig. 5). Similarly, it yielded a lower RMSE (1.04 g/kg) than the traditional methods (RMSE: 1.74~1.88 g/kg). It demonstrated the superior predictive accuracy of the 3D INLA-SPDE approach.

In addition, it is worth noting that a major challenge in fitting traditional digital soil mapping methods is the "large *N* problem", that is, the computational burden would exponentially increase with increase of data size [76], [82], [83]. Compared with traditional prediction methods, the 3D INLA-SPDE method, which is based on the INLA-SPDE model framework and uses sparse matrices, can effectively overcome this problem [61], [76], [84]. It integrates prior knowledge with monitoring data through a Bayesian framework and uses sparse matrix calculations to reduce computational burden, improve computational speed, and relax constraints on computational resources [76], [85]. Therefore, the 3D INLA-SPDE model can enhance computational efficiency while maintaining high accuracy, which is validated by the clearly lower time for model elapse of 3D INLA-SPDE model (about 50 minutes) than 3D RF model (around 80 minutes) in current study.

Accurate uncertainty quantification is another essential issue in spatial prediction modeling. The 3D INLA-SPDE method generates posterior statistical parameters and visualizes the spatial distribution of uncertainty [72]. By generating posterior parameter distributions, the model captures the full range of possible predictions and their probabilities, enabling uncertainty quantification. This represents a significant advantage over the "black-box" nature of many traditional machine learning approaches, let alone to say that traditional deterministic geostatistical methods like 3D IDW cannot provide such probabilistic uncertainty estimates [47], [48], [49]. When using machine learning model like RF, a large number of simulations (usually no less than 100 times) should be made to output the uncertainty of the predictions, which is inefficient and time-consuming [56], [57], [86], [87]. The 95% uncertainty interval produced by the 3D INLA-SPDE model defines the range within which the true value is expected to occur with 95% probability. This facilitates reliable spatial interpretation of prediction results [83].

Overall, this study introduces a key innovation by extending the traditional INLA-SPDE framework from 2D planar prediction to 3D spatial prediction. Conventional 3D geostatistical and hierarchical prediction methods often inadequately quantify uncertainty and capture vertical anisotropy in 3D spatial attributes, leading to limited predictive accuracy [88], [89], [90]. By incorporating depth as a structural dimension and employing the Matérn covariance function, this study addressed these limitations [30]. The 3D INLA-SPDE model successfully captured the consistent decrease in SOC content with increasing depth (Figs. 3 and 6). This represents an improvement over hierarchical methods, which often fail to account for complex vertical dependencies, resulting in reduced accuracy [15], [91]. The Matérn covariance function, central to the INLA-SPDE approach, is particularly suited for modeling spatial dependence in multidimensional data [92], [93]. Its flexible parameters allow customization to capture varying degrees of smoothness and spatial correlation ranges [26], [94]. Treating depth as a structural dimension enables coherent integration of information across depths, leading to more accurate 3D spatial representation. Optimization of mesh parameters (e.g., *cutoff* = 150 m, *maxedge* = 6 km) is essential for balancing predictive accuracy and computational efficiency. The mesh enables 3D spatial discretization within the SPDE framework. Excessively fine meshes incur high computational costs, whereas overly coarse meshes reduce predictive accuracy [30]. Parameter optimization yields a mesh structure that ensures prediction quality while maintaining computational tractability. This balance between computational accuracy and efficiency represents a key advantage of the INLA-SPDE approach.

### *D. Key Drivers of SOC Variation in 3D Domain in Poyang Lake Wetland*

This study employed SHAP-based explainable machine learning to systematically investigate the main drivers of SOC in the Poyang Lake Wetland [69], [95] (Figs. 7, 10). Analysis of variable importance rankings and PDPs elucidated the influence mechanisms of soil properties, climate, topography, and biological factors on SOC dynamics [96] (Fig. 10).

**Fig. 10.** Partial dependency graph of each covariate of SOC based on the SHAP method. Each point represents the marginal contribution of a feature to the local prediction relative to the global SOC average prediction (the Shapley value of the corresponding covariate in g/kg on the Y-axis of the dependency graph, and the value of the corresponding covariate on the X-axis). The red line on the dependency graph is the Shapley value fit line. The correlation test is the examination of each covariate and the corresponding SHAP value of the covariate: * : $p<0.05$; $p<0.01$; *** : $p<0.001$.

As illustrated in Fig. 10, soil attributes contributed 53.7% to the variation of SOC in soil profile of Poyang Lake Wetland, with sand content exhibiting the highest mean absolute SHAP value (0.085), identifying it as the primary factor governing SOC spatial distribution. The PDP reveals that the relationship between sand content and SOC shifts from positive to negative when sand content exceeds 25%. Low sand concentrations facilitate SOC accumulation through their small pore structure and ideal nutrient retention capacity, which promotes microbial activity and organic matter decomposition while enabling synergistic interactions with other soil particles [97]. Higher sand content increases soil porosity and aeration, enhancing aerobic microbial activity and accelerating organic matter decomposition, thereby reducing SOC content [98]. Silt initially disrupts organo-mineral associations, weakening its positive effect on SOC, but subsequently promotes accumulation of SOC by forming stable micro-environments. Clay exerts a modest positive effect through organic carbon adsorption and stabilization, though this effect is limited by its low proportional representation in the soil [99]. Among soil nutrients, AK exhibited a mean absolute SHAP value of 0.066. The PDP demonstrated a significant positive correlation between AK content and SHAP values ($p < 0.001$). AK enhances organic matter input and stabilization indirectly by promoting root growth and microbial activity [100]. Soil pH

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had a mean absolute SHAP value of 0.059. The PDP indicates that the positive effect of pH on SOC peaks within the 5.4~5.6 range. Beyond this range, shifts in microbial community structure may lead to accelerated decomposition rates [101].

Topographic factors contributed to 23.0% of the SOC spatial heterogeneity. The TWI exhibited a mean absolute SHAP value of 0.057, indicating its notable influence. The PDP reveals that in low-lying areas (TWI > 11), TWI exerts a positive effect on SOC. This positive relationship is attributed to long-term flooding conditions, which inhibit aerobic microbial activity [102] and slow SOC decomposition. Slope had a mean absolute SHAP value of 0.046, confirming its significant role in SOC spatial differentiation. However, due to the region's gentle overall relief, the influence of Slope variation was less pronounced. The MrVBF had a mean absolute SHAP value of 0.043. The PDP indicates that areas with MrVBF > 8.0 represent typical depressions or valley bottoms characterized by very flat terrain and large catchment areas. These topographic conditions promote long-term or seasonal waterlogging, resulting in an anaerobic environment. This environment slows the decomposition of organic matter. Furthermore, these stable landforms are more effective at capturing and depositing organic matter, collectively enhancing SOC accumulation [103].

Climate variables accounted for 12.1% of the explained variance in SOC dynamics. MAP exhibited the highest mean absolute SHAP value (0.057) among climate factors. The PDP shows a positive correlation between MAP and SOC, primarily attributed to enhanced plant productivity and increased water input [104]. MAT had a mean absolute SHAP value of 0.046. The PDP indicates a gradual negative relationship between MAT and SOC, although the overall effect size was modest. This limited temperature effect likely results from the constrained MAT variation (<1°C) in the study area. Generally, elevated temperatures enhance microbial activity, accelerating organic matter decomposition and reducing SOCS [105].

Biological factors contributed 11.2% to SOC dynamics, with the NDVI and EVI each exhibiting a mean absolute SHAP value of 0.046. PDPs reveal that the positive effects of NDVI and EVI on SOC gradually weaken with increasing vegetation coverage, becoming slightly negative when NDVI exceeds 0.7 and EVI exceeds 0.5. This nonlinear response occurs because high vegetation coverage enhances autotrophic respiration and accelerates litter decomposition, while simultaneously reducing soil moisture and intensifying nitrogen limitation. These changes collectively inhibit microbial carbon sequestration and promote organic matter mineralization [106]. This pattern highlights a characteristic nonlinear relationship between vegetation density and carbon cycling in wetland ecosystems , [107]

This study demonstrates that SOC dynamics in the Poyang Lake Wetland are governed by the interacting effects of soil properties, climate, topography, and biological factors. Future research could benefit from integrating process-based models with machine learning approaches to better elucidate the biochemical mechanisms driving SOC cycling [108].

### *E. Limitations and Future Research Directions*

As described previously, although our developed 3D INLA-SPDE model demonstrated superior performance in terms of 3D spatial prediction capability, interpretability, uncertainty analysis, and computational efficiency, several limitations of current study should be noted. A primary limitation of this study is the sparsity and uneven spatial distribution of soil samples [109]. This data scarcity likely compromises the accuracy of subsurface SOC predictions, as the model may lack representativeness and fail to fully capture conditions and dynamics in deeper soil layers. Future studies should prioritize collecting more comprehensive soil samples and testing the superiority of INLA-SPDE model with sufficient samples [112].

Secondly, the limitations of the environmental covariates derived from remote sensing data also need to be taken into account. The original spatial resolution of these variables (90 m) is not sufficient to capture the subtle spatial heterogeneity at the micro-level within complex wetland landscapes, and the inherent pixel-point scale mismatch-where each pixel integrates reflectance over a continuous area while a soil sample represents a point-further introduces representativeness errors, thereby potentially missing their driving effect on changes in SOC. Moreover, the depth of information provided by surface remote sensing itself has limitations, making it difficult to directly infer underground soil properties or identify underground processes that affect deep SOC variation.

The potential impact of outliers on spatial mapping also warrants consideration. Although the 3D INLA-SPDE model accounts for uncertainty, extreme values can affect predictions, particularly in sparsely sampled areas [92]. Integrating outlier detection algorithms into the modeling framework could mitigate this issue [110] , thereby enhancing the accuracy and reliability of spatial predictions. Incorporating a temporal dimension represents a promising direction for future research. Finally, by leveraging the fusion of multi-source remote sensing data, the spatiotemporal resolution can be enhanced, to accurately capture the seasonal dynamic changes of vegetation[111]. At the same time, deep learning can be combined to explore the deep features of remote sensing images, and quantify the temporal evolution patterns of SOC, providing a more comprehensive understanding of the SOC cycle [112].

## V. Conclusion

This study developed and applied a novel 3D INLA-SPDE model for mapping 3D variation of SOC in the Poyang Lake Wetland. Our developed 3D INLA-SPDE model successfully captured the 3D spatial distribution patterns and identify driving mechanisms of 3D variation of SOC, providing important scientific support for wetland carbon management. Compared to traditional 3D digital soil mapping methods, the proposed model shows clear advantages in computational feasibility, prediction accuracy, as well as uncertainty quantification. The main features and advantages of the proposed model include:

(1) Our study advances beyond traditional 3D spatial analysis frameworks by extending the INLA-SPDE model to 3D space, enabling precise quantification of SOC spatial heterogeneity. The model demonstrated clearly higher predictive accuracy ($R^2$= 0.80, RMSE= 1.04 g/kg) than traditional geostatistical and machine learning model, with $R^2$ increased by 77.78%~116.22%, the RMSE reduced by 40.23%~44.68%.

(2) By accounting for more sources of uncertainty and

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employing the Matérn covariance function to accurately characterize both vertical attenuation patterns and horizontal heterogeneity of SOC in 3D prediction, the INLA-SPDE model achieves substantially reduction on prediction error. The Bayesian framework provides a comprehensive and easily obtained uncertainty quantification, delivering 95% confidence intervals and a complete spatial uncertainty assessment solution.

(3) The 3D INLA-SPDE model significantly enhances computational efficiency by leveraging sparse matrix operations, and compared with 3D RF, the computational time for 3D INLA-SPDE model reduced by 37.5%.

Overall, here we developed a novel and promising new tool for predicting 3D variation of soil properties which enriched the tools of DSM. However, more efforts are still necessary to confirm the feasibility and superiority of our developed 3D INLA-SPDE model for different soil properties, across different spatial scales and regions under different soil sample size. This study clarifies the 3D variation of SOC in the top 100 cm soil of China's largest freshwater wetland and provides an estimate of its SOCS. It could support the development of sustainable and climate-smart wetland management measures.

## CRediT authorship contribution statement

Bifeng Hu: Writing -review & editing, Sampling, Data analysis, Software, Conceptualization. Yibo Geng: Writing-original draft, Sampling, Methodology, Data analysis, Software, Validation. Hongyi Li: Supervision, Project management, Conceptualization. Zhou Shi: Supervision, Sources. Qian Zou: Data collection and analysis. Jinghua Wang: Data collection. Songchao Chen: Writing -review & editing. Qian Zhu: Sampling. Wenjun Ji: Sources. Modian Xie: Sampling, Writing -original draft, Conceptualization, Writing -review & editing.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have influenced the work reported in this paper.

## Acknowledgements

The author would like to express gratitude to Hanjie Ni, Jiangtao Yang, Yali Wen, Xiangyu He, and Chongcan Wu for their valuable assistance during the field sampling.

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This article has been accepted for publication in IEEE Transactions on Geoscience and Remote Sensing. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TGRS.2026.3737791



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This article has been accepted for publication in IEEE Transactions on Geoscience and Remote Sensing. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TGRS.2026.3737791





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